

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459523.6562 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.022
Transit depth (Rp/Rs)^2	2.05 +/- 0.64 [%]
Orbital Inclination (inc)	89.49 +/- 0.82
Airmass coefficient 1 (a1)	0.999 +/- 0.013
Airmass coefficient 2 (a2)	0.001 +/- 0.023
Scatter in the residuals of the lightcurve fit is	1.29 %
Best Comparison Star	#3 - [398, 105]
Optimal Aperture	3.96
Optimal Annulus	12.76
Transit Duration (day)	0.1106 +/- 0.003

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.143 ± 0.022 vs 0.1446 (deviation 0.07σ)
Duration fit vs published	0.1106 d vs 0.1087 d (deviation 0.62σ)
Mid-time O–C	-8.8 min
Depth SNR	6.5
Residual scatter	1.29 %

Light curve

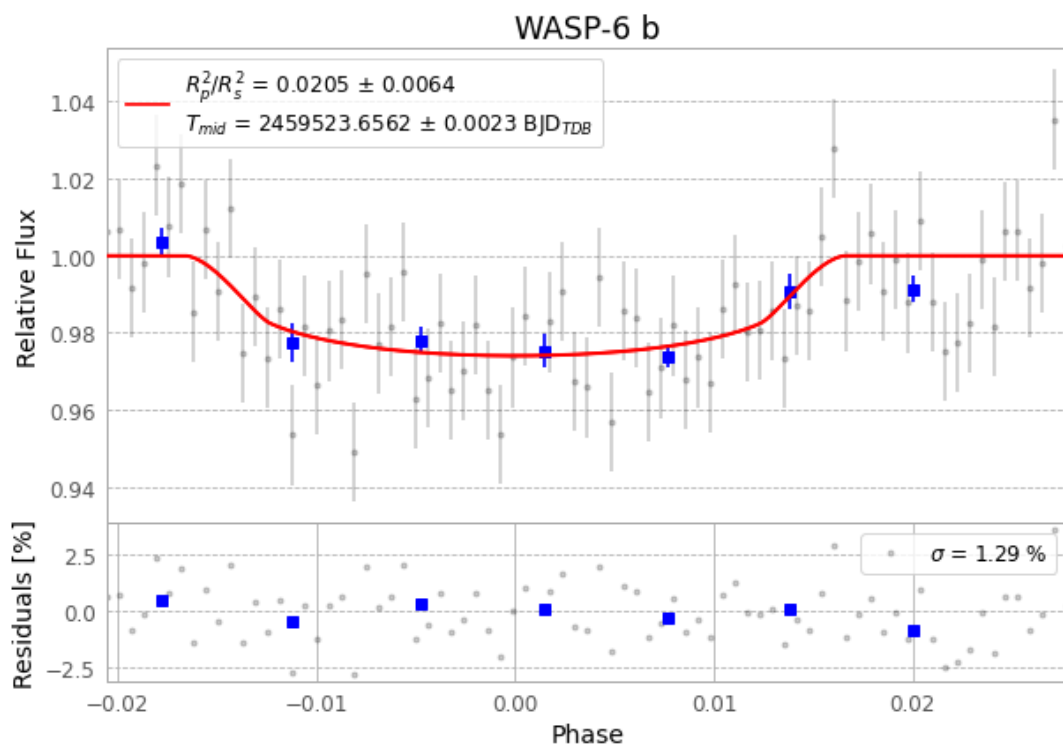


Figure 1 — FinalLightCurve_WASP-6 b_2021-11-04.png (SHA-256 0cfa38355b25c883...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [512, 78], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3a02da770bf16269...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

79 FITS frames (52 MB), WASP-6211105020012.FITS → WASP-6211105055412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-6-211105.log (8e837e9b7046...), FinalLightCurve_WASP-6 b_2021-11-04.png (0cfa38355b25...), FinalLightCurve_WASP-6 b_2021-11-04.csv (441befc618aa...)

Compute resources

Wall time 585.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 00:26Z.